

RoPE Has a Spectral Budget

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Abstract

RoPE has a finite spectral budget: each head must place only $K = d_{\text{rot}}/2$ rotary frequencies before training. The standard geometric table spends this budget uniformly in log-frequency, while long-context methods usually alter or transport its range; we ask whether interior allocation still matters when that range is fixed. In a 151.9M exact-range matched pair, only the 30 interior frequencies differ, and endpoint-normalized Cosh minus the uniform grid changes trained-model OOD NLL by $-0.478/-0.205/-0.113$ at 512/1K/2K. In a separate 12-configuration, three-seed factorial, pre-specified Cosh and matched-exponential allocations beat the uniform grid in 10/12 and 9/12 configurations. We then formulate a convex variational collision surrogate whose unique minimiser is a cosh density; quantising its inverse CDF gives EVQ-COSH, a closed-form schedule with no learned parameters and geometric RoPE as its $\tau \rightarrow 0$ limit. The allocation remains binding at mature scale. On a 1.485B checkpoint, Native and EVQ-COSH receive identical Q/K-only adaptation and identical phase exposure through 16K; their 8K RULER scores are 2.02% and 31.63%. A matched 8B continuation reaches 14.03% versus 0.295% at 16K. Thus range determines where the spectrum lies, while allocation determines how a finite set of channels occupies it.

1. Introduction

RoPE has a spectral budget. Each attention head receives only $K = d_{\text{rot}}/2$ rotary frequency channels, and those channels must be placed before the model sees a token. Standard geometric RoPE (Su et al., 2024) makes a specific allocation: it spaces the channels uniformly in log-frequency, $\omega_i = b^{-2i/d_{\text{rot}}}$. Because this table is inherited across model

families — from LLaMA-3 and Qwen2.5 to DeepSeek-V3 and video DiTs (Grattafiori et al., 2024; Yang et al., 2024; DeepSeek-AI, 2024b; Wei et al., 2025) — its allocation is easily mistaken for part of the rotary operator itself.

Recent work makes the distinction consequential. Positional interpolation, YaRN, and LongRoPE-style search transport a realised spectrum toward a target context (Chen et al., 2023; Peng et al., 2024; Ding et al., 2024; Shang et al., 2025); FMRoPE retargets the geometric band through its base (Oka et al., 2026); LeRoPE instead learns the individual frequencies and finds stable non-geometric tables (Karypis et al., 2026). Range, transport, and learning all show that the default table is not sacrosanct. They leave a sharper question: *at a fixed spectral range, is the allocation of the finite interior channels an independent training-time variable, and can it be chosen in closed form?*

We answer the first part with an exact-range identification experiment (§4.2). The two arms share the sampled highest and lowest frequencies, log-span, model initialisation, token order, optimiser, training budget, and evaluation anchors. Only the $K-2 = 30$ interior frequencies move. After training, endpoint-normalised cosh minus the uniform-in-log grid changes OOD NLL by $-0.4775/-0.2050/-0.1128$ at 512/1K/2K in the seed-42 matched pair. In a separate 12-configuration, three-seed factorial, pre-specified non-uniform allocations are favored in 10/12 and 9/12 configurations. Allocation changes what the model learns even when the range is held fixed.

The second part becomes a variational problem. We represent a frequency table by a density $\rho(\phi)$ over log-frequency and minimise a convex broadband collision surrogate. Its unique constrained minimiser is a cosh density; its inverse CDF yields EVQ-COSH, a one-parameter analytic warp that changes only the RoPE inverse-frequency initialisation (§3). It adds no learned parameters, auxiliary loss, or search, and recovers geometric RoPE as $\tau \rightarrow 0$. A softmax-transport analysis supplies the scaling $\tau \propto d_{\text{eff}}/\sqrt{L_{\text{train}}}$; the finite coefficient is calibrated as an operating point and tested across the factorial.

The effect also survives beyond mechanism-scale controls (§4.3). On OLMo-2 (1.485B), Native and EVQ-COSH receive the same 300-step Q/K-only continuation, data, module budget, and 4K/8K/16K phase exposure; the active

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Table 1. Nonexclusive interventions on $\omega_i = b^{-u_i}$. Methods may act at multiple levels; learned and searched tables differ from EVQ-COSH in how their channel-wise frequencies are obtained.

	Controlled object	Typical stage	Representative methods
1	deployed phases / spectrum	context extension	PI, YaRN, LongRoPE
2	scalar base b	training or inference	FMRoPE
3	channel frequencies $\{\omega_i\}$	before / during training	EVQ-COSH, LeRoPE, AdaRoPE

method variable is the frequency table. At 8K, Native/EVQ-COSH scores are 2.02/31.63% on RULER and 0/17.5% exact on held-out 2WikiMultiHopQA. Matching long-phase exposure therefore does not erase the frequency-substrate effect. On LLaMA-3-8B, a separate matched physical-8K continuation moves 16K RULER from 0.295% to 14.03%. These experiments carry the identified variable to mature checkpoints and autoregressive effective-context tasks.

What this paper establishes.

- **The axis.** Exact-range controls identify interior frequency allocation independently of base and span, with cross-configuration replication (§4.2).
- **The construction.** A convex variational surrogate yields the closed-form, zero-learned-parameter EVQ-COSH schedule and recovers geometric RoPE as a limit (§3).
- **The mature attribution.** Phase-matched 1.485B adaptation and matched 8B adaptation show that the frequency substrate remains consequential for greedy multi-hop QA and RULER-family evaluation (§4.3).

2. Related Work

Write the RoPE table as $\omega_i = b^{-u_i}$, with the geometric grid $u_i = 2i/d_{\text{rot}}$. Prior work changes several, nonexclusive parts of this parameterization: it transports a pretrained spectrum to a longer range, changes the scalar base, or changes the frequencies assigned to individual channels (Table 1). This decomposition separates when a method acts from which spectral degrees of freedom it controls.

Context-range extension. Position interpolation rescales positions to keep rotary phases within the training range (Chen et al., 2023). YaRN, SelfExtend, PoSE, and CLEX refine this extension through frequency-dependent scaling, remapped positions, or extension-time training (Peng et al., 2024; Jin et al., 2024; Zhu et al., 2024; Chen et al., 2024). LongRoPE and LongRoPE2 search channel-wise rescaling factors for pretrained models (Ding et al., 2024; Shang et al., 2025). These methods may combine several intervention levels; their common goal is to transport a learned model to a declared longer context. Bounds based on the RoPE base characterize limits of this regime (Xu et al., 2024).

Base frequency and emergent bands. RoPE models use the geometric spectrum non-uniformly. Barbero et al. identify frequency-localized structure in query and key norms and introduce partial RoPE, which removes the slowest rotations (Barbero et al., 2025). Oka et al. show that the high-norm band’s location depends jointly on the scalar base and the training length (Oka et al., 2026). Their FM-RoPE rule sets $b = L_{\text{train}}$ during training or fine-tuning and may retarget b to the evaluated length at inference. Thus FMRoPE moves the geometric spectrum by changing one scalar while retaining a uniform exponent grid. By contrast, an exact-range allocation test can hold the endpoints and log-frequency span fixed while changing only the interior grid (§4.2).

Learning and constructing frequency tables. LeRoPE learns one log-scale per frequency band, shared across layers and heads, and reports consistent non-geometric profiles on a 52M–2.5B model ladder (Karypis et al., 2026). In its 217M ablation, retraining with a frequency table frozen from an independent LeRoPE run retains 63.6% of the full validation-perplexity gain over RoPE, compared with 10.4% for partial RoPE. This isolates substantial value in the fixed table, although obtaining that table still requires a separate learned run. AdaRoPE instead learns per-head frequencies together with head-specific, length-dependent attention scaling (Wang et al., 2026). Recent analysis further shows that the frequencies used by a trained model depend on the relative-distance structure of its training data, with the preferred frequency varying inversely with dependency width (Wu et al., 2026).

Resonance RoPE analytically snaps each wavelength to a nearby integer period to close phase gaps during interpolation, producing a fixed non-geometric table that can be combined with range scaling (Wang et al., 2024). EVQ-COSH instead derives an interior allocation density from a variational objective and tests that variable with the sampled range held fixed.

EVQ-Cosh. EVQ-COSH addresses a different fixed-table objective: it warps the exponent grid before training using the closed-form solution of a stated variational surrogate. The result is shared across heads, introduces no learned positional parameters, and does not require a deployment target. It therefore complements range extension and differs from LeRoPE and AdaRoPE in how the non-geometric table is obtained, not in whether frequency allocation matters. Alternatives that learn a broader positional function, including DAPE, FIRE, and FoPE (Zheng et al., 2024; Li et al., 2024; Hua et al., 2025), change a larger functional object than a fixed RoPE table.

Evaluation. Long-context evaluation spans retrieval diagnostics and task suites (Bai et al., 2024; Hsieh et al., 2024). Information can be lost despite being present in the prompt (Liu et al., 2024), and positional interventions can mitigate this failure (Zhang et al., 2024). We use PE-dominant diagnostics for mechanism and RULER for effective-context claims, while keeping task-family adaptation distinct from unseen-task transfer.

3. Theory and Method

The minimiser of $\mathcal{C}_{\text{app}}$ is exact conditional on the stated convex surrogate; the pure-tether branch then gives a closed-form modelling choice. A separate post-softmax calculation supplies the $d_{\text{eff}}/\sqrt{L_{\text{train}}}$ scaling structure, while its finite prefactor is selected empirically.

3.1. RoPE as frequency allocation

Let $\phi \in [0, 1]$ parameterise log-frequency, $\omega(\phi) = b^{-\phi}$. Let d_{rot} be the number of RoPE-rotated dimensions and $K = d_{\text{rot}}/2$ the number of rotary pairs. For standard MHA $d_{\text{rot}} = d_{\text{head}}$; for compressed-RoPE architectures such as MLA (§4.5) d_{rot} may be smaller, in which case the inverse-CDF grid uses $K = d_{\text{rot}}/2$. The concrete MLA study uses its pre-specified τ and is not treated as a test of the general operating-point rule. A positive density $\rho \in C^2([0, 1])$ with $\int_0^1 \rho = 1$ induces a discrete allocation by inverse-CDF quantisation, $\phi_k = F_\rho^{-1}(u_k)$, $F_\rho(\phi) = \int_0^\phi \rho$. Geometric RoPE is $\rho \equiv 1$.

3.2. A surrogate collision functional

For a bounded-support separation prior D , the exact phase-collision kernel $K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta) d\Delta$ is oscillatory and is not pointwise approximated by a smooth two-term model. We therefore analyse the surrogate quadratic form

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho(\phi)^2 d\phi + \frac{\beta}{2} \iint_{[0,1]^2} \rho(\phi)\rho(\psi) \min(\phi, \psi) d\phi d\psi, \quad (1)$$

with $\alpha, \beta > 0$ fitted to the exact kernel on a finite channel grid (App. A.12), giving $\alpha \approx 1/d_{\text{rot}}$. The diagonal term penalises channel-load concentration; $\min(\phi, \psi)$ is the Green kernel of $-\partial_\phi^2$ on $[0, 1]$ and penalises pairs of channels that are *both* slow, whose phase features are strongly correlated over a finite separation window. This is a spectral-packing penalty; it does not assume that slow channels carry no content or positional function. $\mathcal{C}_{\text{app}}$ is a discrete-grid surrogate, not the continuum stationary-phase asymptotic of K (App. A.5).

3.3. Constrained stationary allocation

Minimising (1) subject to $\int_0^1 \rho = 1$ with multiplier ν , the first variation gives $\alpha\rho(\phi) + \beta g(\phi) + \nu = 0$ with $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$. Since $g'' = -\rho$, every interior stationary density obeys $\rho'' - \tau^2 \rho = 0$ with $\tau = \sqrt{\beta/\alpha}$; pre-differentiated stationarity and mass normalisation give $\rho'(0) = -\tau^2$, $\rho'(1) = 0$ (App. A).

Theorem 3.1 (Exact stationary allocation under the surrogate). *Under (1) with $\alpha > 0, \beta \geq 0$ (so $\mathcal{C}_{\text{app}}$ is convex) and constraints $\rho \in C^2([0, 1])$, $\rho > 0$, $\int_0^1 \rho = 1$, the unique constrained minimiser is*

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}, \quad (2)$$

strictly positive on $[0, 1]$ for $\beta > 0$; at $\beta = 0$ the minimiser is the $\tau \rightarrow 0$ limit $\rho_0 \equiv 1$ (uniform/geometric).

Here $\tau = \sqrt{\beta/\alpha}$ indexes the exact surrogate family. The deployed value is *not* obtained by equating it to the fitted coefficient ratio; it is selected by the separate operating-point model below.

The forced extension and the choice of the closed-form homogeneous branch are derived in App. A.2.

3.4. Closed form, and the geometric limit

Corollary 3.2 (Closed-form pure-tether quantiles). *For $\tau > 0$ the density of Theorem 3.1 has CDF $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$, so inverse-CDF quantisation gives*

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k+1/2}{K}. \quad (3)$$

As $\tau \rightarrow 0$, $\phi_k(\tau) \rightarrow u_k$, recovering the uniform-log midpoint grid within the pure-tether sub-family.

The complete recipe. For effective head dimension d_{eff} and training length L_{train} :

$$\tau = d_{\text{eff}}/\sqrt{L_{\text{train}}}, \quad K = d_{\text{rot}}/2, \quad u_k = (k + \tfrac{1}{2})/K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (4)$$

Set τ , compute midpoint quantiles, evaluate (3), replace the inverse frequencies. No auxiliary loss, no extra parameters, no optimiser dependence, and no declared deployment length. The endpoint grid $u_k = k/K$ is an analytic anchored variant ($\phi_0 = 0$ preserves $\omega_0 = 1$); the deployed implementation uses midpoint quantisation, and experiments using the anchored variant are labeled separately.

3.5. What the schedule concretely does

At $\tau = 4$, $b = 5 \times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u = 0.5$

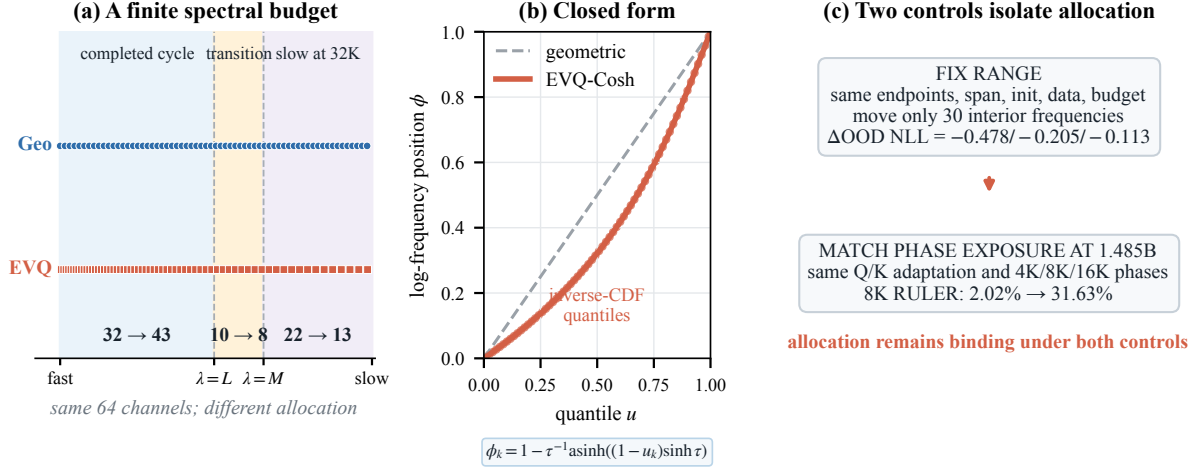


Figure 1. EVQ-COSH frequency allocation. Geometric RoPE places channels uniformly in log-frequency; EVQ-COSH applies a closed-form inverse-CDF warp to reallocate the same number of channels. In the 1.485B geometry, this moves channels toward wavelengths that complete a training-window cycle. Exact-range and phase-matched controls isolate allocation from range and exposure. Rotation, architecture, and optimiser are unchanged; no learned parameters are added.

moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed midpoint grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$ has relative density 0.76. Thus thinning concentrates on channels with $1/\omega \gg L_{\text{train}}$ while preserving more of the training-scale band. This motivates a directional prediction: channels with $1/\omega \ll L_{\text{train}}$ complete many periods during training, so positions beyond L_{train} land in already-trained phase regions, while channels with $1/\omega \gg L_{\text{train}}$ never complete one and extrapolate into unseen phase. Moving budget toward the former should favour extrapolation, while any in-window cost must be measured after training. The surrogate does not assign the much larger mismatch observed when a pretrained model is retrofitted with a new table.

3.6. Where the finite budget moves

The allocation is easiest to read in wavelength, $\lambda(\phi) = 2\pi/\omega(\phi) = 2\pi b^\phi$. For a training span L and an evaluation ceiling $M > L$, the exponent axis has three geometric regions: $\lambda \leq L$ completes at least one cycle during training; $L < \lambda \leq M$ varies across the target window without completing a cycle during training; and $\lambda > M$ completes no cycle even at the target ceiling. These regions describe phase geometry, not a claim that any channel is unused.

Lemma 3.3 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \frac{1}{\tau} \operatorname{arccosh}\left(\frac{\sinh \tau}{\tau}\right) \leq 1 - \frac{1}{\sqrt{3}} \approx 0.4226. \quad (5)$$

Thus EVQ-COSH assigns more density than the geometric grid for $\phi < \phi_c$ and less for $\phi > \phi_c$.

For OLMo-2, $b = 5 \times 10^5$, $L = 4096$, and $M = 32768$ place the two boundaries at $\phi_L = 0.494$ and $\phi_M = 0.652$, both to the right of the universal crossing bound. At $\tau = 2$, the continuous mass in the three regions changes from 0.494/0.158/0.348 under a uniform grid to 0.671/0.122/0.208. The corresponding 64-pair midpoint tables contain 32/10/22 versus 43/8/13 channels. The warp therefore buys more completed-cycle channels by spending density in both longer-wavelength regions. This is an exact budget statement, not a trained-loss theorem; it predicts a measurable reach–resolution trade that the mature-model experiments test. Proof and a complementary divergence inequality are in App. A.9.

3.7. The operating-point rule, and its status

Shape and scale come from different layers. The cosh shape is the pre-softmax stationary solution under $\mathcal{C}_{\text{app}}$; the scale is selected *post-softmax*.

Proposition 3.4 (Semi-analytic operating-point scaling).

On the pure-tether family, assuming $Q_1(L, b) > 0$ and a diffuse post-softmax baseline $p_0=1/L$, the Pearson- χ^2 channel-load stiffness and post-softmax transport utility admit small- τ expansions $S_{\chi^2}(\tau) = c_S \tau^4 / d_{\text{head}} + O(\tau^6)$ with $c_S = \frac{1}{45}$, and $U(\tau, L) = (d_{\text{head}}/L)[Q_0 + \tau^2 Q_1(L, b) + O(\tau^4)]$, with $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$, $\eta(\phi) = (1-\phi)^2/2 - 1/6$, and $q(x) = 1/2 + \sin(2x)/(4x) - (\sin x/x)^2$ (App. A.13). The leading-order stationary point of $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ satisfies

$$\tau_*^2 = 45\lambda Q_1(L, b) \cdot \frac{d_{\text{head}}^2}{L}, \quad \tau_* \propto \frac{d_{\text{head}}}{\sqrt{L}}.$$

Within this model, the explicit $d_{\text{head}}/\sqrt{L}$ factor is modulated by $Q_1(L, b)$; this is a scaling structure, not a pure power law. The prefactor $c(\Pi) = \sqrt{45\lambda Q_1(L, b)}$ remains convention- and protocol-dependent. At practical τ the exact cosh warp is evaluated directly—the small- τ series motivates the scaling and does not certify a finite operating point. For non-diffuse attention, a softmax-Jacobian Rayleigh quotient provides a curvature diagnostic, but connecting that diagnostic to the utility above is an additional modelling assumption (App. A.14).

We therefore deploy the zero-search convention $c=1$ and test it as an operating default. In one direct selection sweep, selected $\tau=5$ and the rule value 5.657 differ by 0.0119 selection NLL. Under the stricter exact-range factorial the rule is the best pre-specified Cosh multiplier in 4/12 structural configurations ($0.75 \times / 1.25 \times$ win 2/12 and 6/12). These results support a useful basin initialisation, not point optimality; §4.4 reports the full controlled comparison.

4. Experiments

4.1. Setup and endpoints

We test a sequence of increasingly demanding questions: whether allocation is identifiable at fixed spectral range; whether the effect survives from 50.9M to 8B; which parts of the closed-form prescription the data support; and whether the same range operator behaves differently on the resulting substrates. NLL/PPL, teacher-forced NLL-gap retrieval, causal source use, strict autoregressive exact match, task F1/exact match, and RULER macro remain separate endpoints throughout. EVQ-COSH changes only the training-time frequency table and adds no learned parameters.

The controlled from-scratch runs use fixed GPT-style decoder stacks (Radford et al., 2019), with per-experiment corpora and hyperparameters in App. H. Seed scope is reported with each result.

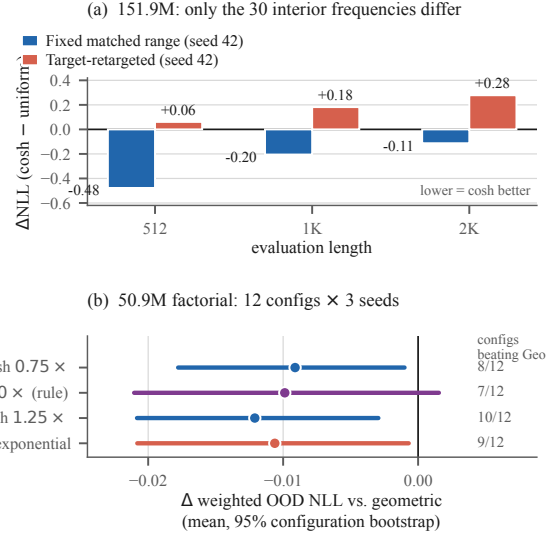


Figure 2. Allocation is identifiable. (a) With sampled extrema and log-span pinned, moving only the 30 interior frequencies shifts trained-model OOD NLL (blue, seed 42); the same contrast reverses once *both* grids are target-retargeted (red). Seed-42 anchors under fixed range: 32/32, 27/32, 22/32; under retargeting, 5/32, 1/32, 2/32. (b) In a separate three-seed, 12-configuration factorial, the $1.25 \times$ Cosh and matched-exponential arms favor non-uniform allocation in 10/12 and 9/12 configurations; the formula point ($1.00 \times$, purple) is not the best of its own family.

Range-scaling arm. All arms labelled +RAMP use our fixed-index smooth-ramp scaler from the NTK-by-parts / YaRN family (Peng et al., 2024). The same implementation and scale are applied to Geo and EVQ-COSH, isolating substrate-dependent leverage of that fixed operator. It is not an official YaRN reproduction or a jointly tuned range-method comparison; the exact rule is given in App. D.

4.2. Identifying the allocation axis

Question. Is training-time allocation separable from base and range selection at all, or is the observed effect a reparameterisation of scalar range quantities?

Protocol. In the seed-42 pair, two 151.9M models share architecture, trainable initialisation, token order, optimiser, LR schedule, batch geometry, a 499,974,144-token budget, and 32 frozen evaluation anchors. The baseline is our paper-faithful implementation of the uniform-in-log rule in Oka et al. (2026, §6.1), using training base 256; the intervention is endpoint-normalised cosh spacing at $\tau=4$. Their sampled extrema and log-frequency span are identical, so only the $K-2 = 30$ interior frequencies move. We measure paired final-128-token teacher-forced NLL.

Result. At the fixed training range, cosh-minus-uniform NLL is $-0.478 / -0.205 / -0.113$ at 512/1K/2K in the seed-42 arm (Fig. 2a). Since every sampled scalar range

Table 2. Exact-range factorial (50.9M, 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). Twelve additional boundary- τ arms are discussed in §4.4. All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times /4 \times /8 \times$, paired within configuration and seed, equally weighted over configurations. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments and is not adjusted across arm contrasts.

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
Cosh $0.75 \times$ – Geo	-0.00912	[-0.018, -0.001]	8/12	0.082
Cosh rule – Geo	-0.00988	[-0.021, 0.002]	7/12	0.125
Cosh $1.25 \times$ – Geo	-0.01210	[-0.021, -0.003]	10/12	0.027
Matched exponential – Geo	-0.01062	[-0.021, -0.001]	9/12	0.071
<i>Cosh and the matched exponential are not empirically separated:</i>				
Cosh (rule) – exponential	+0.00074	[-0.006, 0.008]	7/12	0.836

quantity is pinned, the effect cannot be attributed to base or span selection: *fixed range + different interior allocation* $\Rightarrow$ *different trained-model behaviour*.

Cross-configuration replication. A pre-specified 50.9M factorial spans two bases, two training lengths, three head dimensions, and three paired seeds, with additional boundary- τ arms. It repeats the exact-range control and scores weighted teacher-forced OOD NLL over $2 \times /4 \times /8 \times$, using the 12 structural configurations as the statistical units (Table 2). The pre-specified $1.25 \times$ cosh and matched-exponential schedules favour the non-uniform allocation in 10/12 and 9/12 configurations. The factorial establishes cross-configuration direction; the 151.9M pair provides the direct effect size.

Range and allocation are different controls. Retargeting *both* grids to each evaluation length reverses the seed-42 ordering (Fig. 2a, red). Fixed-range allocation and target-aware range selection therefore answer different questions: the former identifies the interior-allocation axis; the latter can dominate once range is allowed to track the target.

4.3. Phase-matched attribution at mature scale

Question. Once long-range phase exposure is held constant, does the frequency table remain a binding variable on 1B–8B checkpoints and autoregressive effective-context tasks?

1.485B: matching phase exposure does not erase the gap. On OLMo-2-0425-1B-Instruct (OLMo Team, 2025), Native and EVQ-COSH start from their matched Stage-A parents, freeze inherited V/O LoRA tensors, and continue only Q/K LoRA for 300 steps. Both arms use the same data, order, optimiser, trainable parameter count, and 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure; physical sequences remain at most 4K. The active method variable is the fixed frequency table. On all 13 RULER families (Hsieh et al., 2024), Native/EVQ-COSH official macro is 72.19/42.44% at 4K, 2.02/31.63% at 8K, and

0.38/5.03% at 16K. Native receives the same long-phase exposure yet remains at 2.02% at 8K: exposure alone cannot explain the $15.7 \times$ score gap.

1.485B held-out multi-hop QA. The same Q/K-only design, trained independently for 2WikiMultiHopQA, gives Native/EVQ-COSH exact match of 22.0/21.5% at 4K, 0/17.5% at 8K, and 0/4.0% at 16K on 200 held-out examples per length (Bai et al., 2024); token F1 is 25.99/24.84%, 0.07/21.48%, and 0/8.57%. Thus the two substrates acquire nearly identical in-window QA while only EVQ-COSH carries that capability to the longer positions. The prompts use the official questions and contexts with deterministic, answer-filtered distractor filling to realise each length.

Independent 1.485B generation and language modelling. A matched rank-64 Q/K/V/O continuation (Hu et al., 2022) moves 8K niah_single_1 strict first-generated-number exact from 0/100 to 69/100; an independently trained EVQ-COSH seed gives 67/100. On a disjoint set where every source-to-generation gap exceeds the largest training gap, the scores are 0/100 versus 49/100 and 48/100. The same matched seed changes natural-text NLL from 3.735/4.851 to 2.703/2.925 at 8/16K. Separately, a same-initialisation step-0 $\rightarrow$ 1,000 trajectory over 2.097B tokens yields EVQ-COSH-minus-Geo NLL of +0.072/+0.038/−0.044/−0.135 at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K.

8B causal source use and RULER. On LLaMA-3-8B-Instruct (Grattafiori et al., 2024), matched seed-42 ordinary-LM adaptation gives EVQ-COSH-minus-Native-LoRA NLL of +0.390/−1.510/−2.048 at 8/16/32K. On ten frozen true-16K retrieval cases from this protocol, median target-block hit@16 over 32 heads rises from 18.75% to 64.06%; deleting the remote gold block changes NLL by −0.0095 for Native and +1.5055 for EVQ-COSH, establishing causal source use in this diagnostic. A separate identical physical-8K, 13-family adaptation yields official macro 94.44/77.60% at 8K and 0.295/14.03% at 16K; normalized exact is 17.69/21.54% and 0/1.54%, respectively (App. Table 13).

Scope. The 2Wiki and RULER rows are disjoint from training but use the same task families, so they measure task-adapted length transfer. Each matched mature-scale protocol uses one training seed per arm; the second OLMo EVQ-COSH seed replicates only strict NIAH. The 8B deletion study and RULER continuation are separate protocols.

Table 3. Mature-scale effective-context results. Columns are relative to each model’s physical training cap (4K for OLMo-2, 8K for LLaMA-3-8B). OLMo Native and EVQ-COSH receive identical Q/K-only adaptation and phase exposure; LLaMA receives identical Q/K/V/O adaptation. RULER is official macro and 2Wiki is greedy normalized exact. One matched training seed is used per arm; the parenthetical NIAH result is an independent EVQ-COSH training-seed replication.

Model / protocol	Endpoint and arm	1×	2×	4×
OLMo-2 1.485B, selective Q/K	RULER: Native	72.19%	2.02%	0.38%
	RULER: EVQ-COSH	42.44%	31.63%	5.03%
	2Wiki exact: Native	22.0%	0%	0%
	2Wiki exact: EVQ-COSH	21.5%	17.5%	4.0%
OLMo-2 1.485B, Q/K/V/O	NIAH first-number exact: Native	—	0/100	—
	NIAH first-number exact: EVQ-COSH	—	69/100 (67/100)	—
LLaMA-3-8B, Q/K/V/O	RULER: Native	94.44%	0.295%	—
	RULER: EVQ-COSH	77.60%	14.03%	—

4.4. Attributing the schedule and operating rule

Question. Do the trained-model gains identify cosh or the formula value of τ , rather than the broader allocation axis?

The rule selects a basin, not a point optimum. Across the 12 factorial configurations, the formula point is the best of the three pre-specified cosh multipliers in 4; $1.25\times$ wins 6 and $0.75\times$ wins 2. At the pre-specified boundaries, formula $\tau=1$ prefers $1.5\times$, whereas formula $\tau=8$ prefers $0.75\times$. An independent sweep selects $\tau=5$, with the rule value 5.657 only 0.0119 NLL away. Thus $\tau=d_{\text{eff}}/\sqrt{L_{\text{train}}}$ is a search-free basin selector, while the finite coefficient is empirical.

Cosh is a derived instance, not the only effective shape. The exact-range deformation-matched exponential improves over Geo in 9/12 configurations and is not separated from rule-cosh: cosh-minus-exponential is $+0.00074$ NLL, bootstrap $[-0.0055, 0.0075]$, exact sign-flip $p=0.836$. The empirical result identifies interior allocation. Cosh contributes a closed-form, zero-parameter solution of the stated convex surrogate, not a uniqueness claim for trained objectives. The learned-frequency comparator and its tuning budget appear in App. E; shape attribution rests on the fixed schedules above.

4.5. Composition under range scaling and channel scarcity

Same operator, different substrate. We next ask whether a fixed range operator has substrate-dependent leverage. On a 454M model with $L_{\text{train}}=2048$ and a 10% passkey mix (3 seeds), EVQ-COSH + RAMP reaches $100\pm 0\%$ teacher-forced NLL-gap retrieval at 8K against $61\pm 3\%$ for Geo + RAMP. At 12K/16K the scores are 79/68% versus 59/51%, and PPL is 70.9/107.5 versus 82.9/157.7 (Table 4). This is substrate-dependent leverage of the tested fixed $s=8$ operator. Karypis et al. (2026) independently report the same qualitative interaction for a learned non-geometric table.

Table 4. Substrate $\times$ range composition (454M, $L_{\text{train}}=2048$, 10% passkey mix, 3 seeds). RAMP is our fixed-index smooth-ramp scaler (§4.1), applied at the *same* fixed scale $s=8$ to both substrates. PK is teacher-forced NLL-gap retrieval; PK@8K is mean $\pm$ std, other entries are 3-seed means.

Method	PK@8K	PK@12K	PK@16K	PPL@8K	PPL@16K
Geo	41 $\pm$ 5%	57%	51%	161.9	253.2
Geo+RAMP	61 $\pm$ 3%	59%	51%	82.9	157.7
EVQ-COSH	53 $\pm$ 8%	63%	50%	150.3	229.5
EVQ-COSH+RAMP	100$\pm$0%	79%	68%	70.9	107.5

Scarce-channel MLA. MLA (DeepSeek-AI, 2024a;b) provides a direct test with few rotary frequencies. The 432M configuration uses $d_{\text{head}}=64$, $d_{\text{rot}}=32$ (16 frequency pairs), $d_{\text{nope}}=32$, and KV rank 256; it trains at $L_{\text{train}}=8192$ for 500M tokens with learning rate 2×10^{-4} . Across three seeds, the pre-specified $\tau=1.414$ schedule changes $2\times$ extrapolation PPL from 138.8 to 95.6 (-31.1%), with $+0.9\%$ at the training length (App. G.1). Thus the allocation effect persists when only 16 rotary frequency pairs are available.

5. Discussion

RoPE assigns a finite number of rotary channels to a continuous spectral range. The exact-range experiment shows that this assignment is not fixed by the operator, its base, or its sampled endpoints: moving only the interior channels changes the trained model. Frequency allocation is therefore a design decision, not an implementation detail. EVQ-COSH supplies one closed-form, zero-learned-parameter allocation derived from a stated variational objective.

This view also clarifies where allocation enters the modeling pipeline. Allocation shapes the frequency table on which a model is trained or adapted; range transport acts on that table at inference time. Their effects need not be interchangeable. The fixed-scaler and MLA results show that the substrate still matters after a range operator is applied and when only a small rotary subspace is available, while the mature-model results show that the same choice survives beyond scratch-model diagnostics.

The phase-matched OLMo experiment separates allocation from phase exposure. Both arms receive the same target-8K/16K position exposure, continuation data, update budget, and trainable modules, yet Native reaches only 2.02% RULER at 8K against 31.63% for EVQ-COSH. In this matched protocol, long-phase exposure is therefore not a substitute for the table on which those phases are learned. The near tie on 4K 2Wiki exact (22.0/21.5%) also shows that the in-window trade is task-dependent rather than a uniform loss of capability.

The budget accounting in §3.6 makes that trade concrete. For the 1.485B configuration at $\tau = 2$, EVQ-COSH moves 11 of 64 channels into wavelengths that complete a training-window cycle, while the intermediate $L < \lambda \leq M$ region changes from 10 to 8 channels. The family-level RULER scores show a corresponding structure: in a post-hoc split restricted to the explicitly named NIAH families, the three single-needle families change by -21.7 percentage points on average at 4K, whereas the five multi-key/value/query families change by -52.3 points. This does not identify individual channels, but it gives a falsifiable prediction: allocations that thin the intermediate-wavelength region should pay more on simultaneous multi-reference resolution than on single-reference retrieval.

The theory contributes the allocation family and the $d_{\text{eff}}/\sqrt{L_{\text{train}}}$ scaling structure; the finite coefficient is an operating choice inside an empirically broad basin rather than a trained-loss optimum. This separation is useful beyond EVQ-COSH itself: operator design, spectral allocation, and deployment-time transport can be studied as distinct, composable parts of a positional-encoding system. The central implication is that a RoPE table spends a finite spectral budget, and its interior placement should be chosen deliberately before that budget is transported to a new range.

Impact Statement

This work introduces no new dataset, deployed system, or human-subject component. It changes only the initialization of the RoPE inverse-frequency table and adds no learned parameters, so it is inexpensive to audit, ablate, or revert.

A better allocated frequency table may reduce the adaptation or inference effort required for a target context range. The same mechanism could also be used to extend the context capabilities of language or video models, with the general risks associated with stronger generative models. Our evaluations measure positional and long-context endpoints, not factual reliability or safety; deployment claims require application-specific evaluation.

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A. Proof Details

A.1. Collision-only surrogate stationary density

This subsection proves Theorem 3.1 directly from the collision-only surrogate functional; the forced Fisher extension is treated separately in §A.2. Starting from the formal surrogate kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (6)$$

the main-text collision functional is (1), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha \rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (7)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e.}, \int_0^1 \rho = 1\}$. The $\alpha \|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_{\phi}^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (8)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (9)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (10)$$

since differentiating $\alpha \rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha \rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha \rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (9) and (10) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_{\tau}(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (11)$$

which is main-text Theorem 3.1. The mass constraint $\int \rho=1$ is not lost: it enters the boundary condition at $\phi=0$ via $g'(0)=\int_0^1 \rho=1$. Positivity holds on the closed interval since $\rho_{\tau}(\phi) \geq \rho_{\tau}(1) = \tau / \sinh \tau > 0$ for $\tau > 0$ (with $\rho_{\tau}(0) = \tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler–Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (11).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (12)$$

Thus the stationary density obtained above is the unique constrained *minimizer* (not merely a critical point), and the surrogate is well-defined as a quadratic functional on $L^2([0, 1]) \supset C^2([0, 1])$ with no further regularity assumption needed for the δ -component. Conditional on $\mathcal{C}_{\text{app}}$, this derivation is exact.

A.2. Forced Fisher extension

Adding a Fisher utility $-\mu \int \rho(\phi) b^{-2\phi} d\phi$ to $\mathcal{C}_{\text{app}}$ modifies the constrained stationarity to $\alpha\rho + \beta g - \mu b^{-2\phi} + \nu = 0$, whose twice-differentiated form is

$$\rho''(\phi) - \tau^2 \rho(\phi) = \gamma b^{-2\phi}, \quad (13)$$

with γ set by the Fisher multiplier's sign convention. The non-resonant general solution ($\tau \neq 2 \log b$) is

$$\rho_{\text{full}}(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi) + \frac{\gamma}{(2 \log b)^2 - \tau^2} b^{-2\phi}. \quad (14)$$

At resonance, $\tau = 2 \log b$, the forcing profile $b^{-2\phi}$ coincides with one homogeneous mode. The particular solution is then

$$\delta\rho_F^{\text{part}}(\phi) = -\frac{\gamma}{2\tau} \phi e^{-\tau\phi}, \quad (15)$$

with the homogeneous constants again fixed by the boundary and mass conditions. All bounds below containing $(s^2 - \tau^2)^{-1}$ are non-resonant bounds and should be read with $|s^2 - \tau^2|^{-1}$; they are not uniform as $\tau \rightarrow 2 \log b$. EVQ-Cosh is the $\gamma=0$ homogeneous branch. The forced term is a controlled but nonzero residual; its magnitude and the corresponding warp-amplification caveat are bounded in §A.15.

A.3. Closed-form solution and geometric limit

The non-resonant ODE has the closed-form solution

$$\rho^*(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi) + P b^{-2\phi}. \quad (16)$$

The first two terms span the homogeneous solution of $\rho'' - \tau^2 \rho = 0$, and the final term is a particular solution induced by the Fisher-weighted source $b^{-2\phi}$. This decomposition is the source of the main paper's "tether + pulse" interpretation.

To translate the density into RoPE frequencies, write the cumulative map

$$F(\phi) = \int_0^\phi \rho^*(s) ds, \quad (17)$$

normalize it to $[0, 1]$, and invert at the discrete quantiles $u_k = k/K$. In the pure-tether branch this inversion has the closed form

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau). \quad (18)$$

To recover geometric RoPE, expand $\sinh \tau \approx \tau$ as $\tau \rightarrow 0$. Then

$$\phi_k(\tau) \rightarrow 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k)\tau) \rightarrow u_k, \quad (19)$$

which is exactly the uniform quantile map.

This also shows why the high-frequency end remains effectively anchored: for $u_0 = 0$, one has $\phi_0(\tau) = 0$ exactly, so the leading inverse frequency remains $b^0 = 1$, matching geometric RoPE.

A.4. Why the pure-tether branch

The full stationary density contains three terms: a hyperbolic tether ($\cosh, \sinh$) and a particular solution $Pb^{-2\phi}$. The practical EVQ method uses only the pure-tether branch, justified on two grounds.

(1) Closed-form invertibility. The pure-tether density $\rho(\phi) \propto \cosh(\tau(1 - \phi))$ admits an analytically invertible CDF, yielding the closed-form warp $\phi_k(\tau)$. Including the $b^{-2\phi}$ term destroys this invertibility and requires numerical root-finding.

(2) Conditional CDF control. Under the slowly varying Fisher-coefficient assumption in Appendix A.15, the forcing profile is confined to a boundary layer: $\|q_s\|_1 = (1 - e^{-s})/s = O(1/\log b)$. This yields an $O(1/\log b)$ L^1 /CDF residual, not a pointwise bound. Inverse-CDF inversion can further amplify that residual by $1/\inf \rho_\tau \sim \sinh \tau/\tau$. Thus the pure-tether branch is adopted for its closed form; the residual analysis states when that choice is controlled and does not identify a trained-loss optimum.

A.5. Why constant α (and not the continuum stationary-phase coefficient)

A stationary-phase treatment of the exact kernel gives a ϕ -dependent diagonal coefficient rather than the constant coefficient used in $\mathcal{C}_{\text{app}}$. With $\lambda_b := \ln b$, its leading form is $\alpha_{\text{sp}}(\phi) = (\pi D_0 / L \lambda_b) b^\phi$. Writing the finite-grid coefficient as $\alpha_K := 1/(2K)$, the discrepancy is

$$\int_0^1 \rho^2(\phi) [\alpha_K - \alpha_{\text{sp}}(\phi)] d\phi = \frac{\|\rho\|_2^2}{2K} - \frac{\pi D_0}{L \lambda_b} \int_0^1 \rho^2(\phi) b^\phi d\phi, \quad (20)$$

which is not a higher-order correction at the deployed finite grids. Replacing α by $\alpha_{\text{sp}}(\phi)$ produces a different variable-coefficient Euler–Lagrange equation. We use constant α because it is the finite-grid surrogate that yields a positive density with an elementary inverse CDF; this is a modelling decision, not a continuum asymptotic claim. Its empirical check is therefore functional: whether the resulting schedule also reduces the separately evaluated exact-kernel collision score.

A.6. Surrogate quality: functional validation

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ is the primary approximation behind the closed-form density; selecting the pure-tether branch and a finite operating point are additional modeling choices. We test the surrogate functionally rather than pointwise against the oscillatory exact kernel $K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta) d\Delta$: does the derived allocation also reduce the collision objective under K ? This checks the direction selected by the homogeneous branch; it does not bound the omitted Fisher-forcing residual.

Table 5 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute the exact kernel K at the discrete allocation points for both geometric and EVQ, then evaluate the normalized collision score $C = \sum_{i < j} K_{ij}^2 / (K_{ii} K_{jj})$ and effective spectral rank (exponential of the Shannon entropy of the eigenvalue distribution). These diagnostics use the inclusive grid $u_k = k/(K-1)$; they test the allocation direction but are not a direct validation of the deployed midpoint grid. The 12 reported configurations comprise a six-length text sweep at fixed $b=500\text{K}$ and $K=32$, two additional text bases at $L=2048$, and four video bases at $L=32$ and $K=16$; this is not a Cartesian sweep over every listed factor.

Table 5. Functional surrogate check on the inclusive $u_k = k/(K-1)$ diagnostic grid. EVQ reduces collision and increases effective rank under the exact kernel K in all 12 tested configurations. This is a direction check, not a pointwise approximation or a deployed-grid identity test.

Configuration	Collision C		Eff. rank	
	GEO	EVQ (Δ_C)	GEO	EVQ
<i>Text: $K=32$, $b=500\text{K}$, sweep L</i>				
$L=128$	226.3	17.7 (−92%)	2.5	17.0
$L=256$	192.6	27.6 (−86%)	3.0	13.9
$L=512$	163.2	43.3 (−73%)	3.6	10.8
$L=1024$	133.9	59.0 (−56%)	4.4	8.8
$L=2048$	109.0	67.5 (−38%)	5.4	8.0
$L=4096$	87.0	66.3 (−24%)	6.6	8.1
<i>Text: $K=32$, $L=2048$, sweep b</i>				
$b=10\text{K}$	34.8	19.2 (−45%)	13.0	17.3
$b=100\text{K}$	79.4	47.0 (−41%)	7.1	10.4
<i>Video: $K=16$, $L=32$, sweep b</i>				
$b=100$	22.6	14.2 (−37%)	4.8	6.4
$b=1\text{K}$	42.3	28.0 (−34%)	2.8	3.9
$b=10\text{K}$	55.4	39.8 (−28%)	2.2	2.9
$b=50\text{K}$	63.3	47.2 (−26%)	2.0	2.5

EVQ reduces collision under the exact kernel in all 12 configurations (24%–92%), and effective rank increases in every row. In the fixed-base text sweep, the relative reduction grows as L shortens. The separate base sweeps remain positive but are not monotone in b .

A.7. Exact higher-order displacement floor

The one-channel mid-displacement threshold admits an exact Taylor series in $1/K$ that is fully closed-form (no empirical calibration).

Proposition A.1 (Higher-order floor). *For the EVQ-Cosh mapping with $K = d_{\text{rot}}/2$ channels, as $N/K \rightarrow 0$ the N -channel mid-displacement threshold satisfies*

$$\tau_{\text{floor}}(N, K) = 4\sqrt{\frac{N}{K}} \left[1 + \frac{N}{2K} + \frac{241}{120} \left(\frac{N}{K} \right)^2 + O((N/K)^3) \right]. \quad (21)$$

All coefficients are derived from the Taylor series $\Delta\varphi(1/2, \tau) = -\tau^2/16 + \tau^4/256 + 17\tau^6/30720 + O(\tau^8)$, whose coefficients $(-1/16, +1/256, +17/30720)$ are exact rationals from the expansion of $\text{arcsinh}(\sinh \tau/2)/\tau$ (Appendix script `theory_B-floor-higher-order.py`; verified symbolically).

Accuracy. For $N=1$, against the exact floor obtained by solving $|\Delta\varphi(1/2, \tau)| = 1/K$, the leading term has relative error 3.85% at $K=16$ and 0.10% at $K=512$; including both displayed correction terms reduces the error to 0.088% at $K=16$, $1.56 \times 10^{-4}\%$ at $K=128$, and $2.41 \times 10^{-6}\%$ at $K=512$. All constants come from the cosh geometry.

A.8. Single-crossing budget shift

Proof of Lemma 3.3. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$ is strictly decreasing on $[0, 1]$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau/\sinh \tau < 1$. Hence the crossing is unique, and solving $\rho_\tau(\phi_c) = 1$ gives (5). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n+1$. Monotonicity of arcosh yields $\text{arcosh}(\sinh \tau/\tau) \geq \tau/\sqrt{3}$, proving the bound. The three-region masses follow directly from $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$ at $\phi_L = \log(L/2\pi)/\log b$ and $\phi_M = \log(M/2\pi)/\log b$. $\square$

A.9. Waterbed envelope and corollary

Setup. With $\rho > 0$, $\int_0^1 \rho = 1$, $U(d\phi) = d\phi$, $P_\rho(d\phi) = \rho d\phi$, $\delta\rho = \rho - 1$, and $M_\rho(s) = \int_s^1 \delta\rho du$, define the waterbed functional

$$\mathcal{W}(\rho) := \alpha \chi^2(U\|P_\rho) + \beta \int_0^1 M_\rho(s)^2 ds, \quad \chi^2(U\|P_\rho) = \int_0^1 \frac{(1-\rho)^2}{\rho} d\phi. \quad (22)$$

The first term is the Pearson divergence of U from P_ρ (equivalently, $\int 1/\rho - 1$); the second is the integrated squared cumulative error, which arises from the min-kernel off-diagonal covariance in K_{app} .

Proposition A.2 (Waterbed inequality). *Let $D_B(\rho) = -\int_0^1 \log \rho(\phi) d\phi$ be the Burg entropy deficit. Then*

$$\mathcal{W}(\rho) \geq \alpha F_*(D_B(\rho)), \quad F_*(d) = \inf_{\rho > 0, \int \rho = 1, -\int \log \rho = d} \left[\int_0^1 \frac{1}{\rho} - 1 \right], \quad (23)$$

with equality $\mathcal{W}(\rho) = 0$ if and only if $\rho \equiv 1$ (geometric RoPE). A readable corollary via Rényi monotonicity is

$$\mathcal{W}(\rho) \geq \alpha(e^{D_B(\rho)} - 1). \quad (24)$$

Proof sketch. Since $\beta \int M_\rho^2 \geq 0$, $\mathcal{W}(\rho) \geq \alpha \chi^2(U\|P_\rho)$. For fixed $D_B(\rho) = d$, the sharp lower envelope of χ^2 over all densities is the Csiszár–Vajda f -divergence comparison F_* , whose extremizer is a two-level density $X = 1/\rho \in \{a, c\}$ with mass-normalization and entropy-deficit constraints. For the corollary, $1 + \chi^2(U\|P_\rho) = \int 1/\rho \mathbb{E}_U[\cdot] \geq e^{D_1(U\|P_\rho)} = e^{D_B(\rho)}$ by Rényi-2 $\geq$ Rényi-1. Equality at $\rho \equiv 1$ is immediate from both sides vanishing. $\square$

EVQ evaluation. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$ (Eq. (26) below uses the un-normalized integral; cf. §A.16 “Normalization convention”),

$$D_B(\rho_\tau) = \log \frac{\sinh \tau}{\tau} - \frac{1}{\tau} \int_0^\tau \log \cosh y dy = \frac{\tau^4}{90} - \frac{8\tau^6}{2835} + O(\tau^8), \quad (25)$$

$$S_{\chi^2}(\rho_\tau) = \frac{\sinh \tau}{\tau^2} \arctan(\sinh \tau) - 1 = \frac{\tau^4}{45} - \frac{2\tau^6}{315} + O(\tau^8), \quad (26)$$

$$\int_0^1 M_{\rho_\tau}(s)^2 ds = \frac{2\tau^4}{945} + O(\tau^6). \quad (27)$$

Combining,

$$\mathcal{W}(\rho_\tau) = \left(\frac{\alpha}{45} + \frac{2\beta}{945} \right) \tau^4 + O(\tau^6), \quad \mathcal{W}(\rho_{\tau_*}) = O\left(\frac{d_{\text{head}}^4}{(L_{\text{eff}}^J)^2} \right). \quad (28)$$

The centered companion functional is $O(\tau^4)$. It is not the value difference of $\mathcal{C}_{\text{app}}$ itself and does not determine trained-model NLL.

A.10. Surrogate self-consistency theorem

The surrogate $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi)\rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$ is the normalized cosh density.

Theorem A.3 (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (29)$$

Proof. The Green’s function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (30)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (31)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (29). $\square$

An immediate corollary follows.

Corollary A.4 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (32)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2\sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity is an exact internal check on the two surrogate energy terms; it makes no trained-loss claim.

A.11. From scaling structure to an operating default

The post-softmax model uses $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ with $S_{\chi^2}(\tau) = \tau^4/(45d_{\text{head}}) + O(\tau^6)$ and $U(\tau, L) = (d_{\text{head}}/L)[Q_0 + \tau^2 Q_1(L, b) + O(\tau^4)]$. Its leading stationary balance is

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}. \quad (33)$$

Thus the model supplies the powers of d_{head} and L ; the finite coefficient $c(\Pi) = \sqrt{45\lambda Q_1}$ depends on the chosen utility, base, and protocol. Setting $c=1$ is the deployed zero-search convention, not an independently derived constant.

Here

$$q(x) = \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x}\right)^2 = \text{Var}_{t \sim U[0,1]}[\cos(xt)]$$

and $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$. This is a specified phase-variance utility. It should not be identified with the exact oscillatory collision objective or the trained-transformer loss.

The finite- τ evidence is deliberately empirical. A staged 99-run study contains nine small-model configurations and 27 formula-versus-Geo seed pairs, not 27 independent configurations. The default beats midpoint-Geo in 7/9 configuration means and 18/27 pairs, but beats a pilot-selected neighbouring value in only 3/9 held-out configuration means. In the separate exact-range factorial, the formula point is the best pre-specified Cosh multiplier in 4/12 structural configurations. These audits establish the correct role of $c=1$: a useful, fallible basin initialisation.

A.12. Scaling analysis of τ^* : from surrogate to operating point

Numerical characterization of surrogate coefficients. We fit $K_{\text{app}}(\phi_i, \phi_j) = \alpha \delta_{ij} / \Delta\phi + \beta \min(\phi_i, \phi_j)$ to the exact collision kernel $K(\phi_i, \phi_j) = \frac{1}{2L} \left[\frac{\sin((\omega_1 - \omega_2)L)}{\omega_1 - \omega_2} + \frac{\sin((\omega_1 + \omega_2)L)}{\omega_1 + \omega_2} \right]$ evaluated at $K = d_{\text{head}}/2$ uniformly spaced channel positions in $[0, 1]$. Sweeping across $L \in \{128, \dots, 4096\}$, $d_{\text{head}} \in \{32, 64, 128, 256\}$, and $b \in \{10K, 100K, 500K\}$ yields:

$$\alpha \approx \frac{1}{2K} = \frac{1}{d_{\text{head}}}, \quad \beta \sim L^{-0.22}, O(1). \quad (34)$$

The diagonal coefficient α is set by the kernel's diagonal value ($K(\phi, \phi) \approx 1/2$ for most channels) times the channel spacing $\Delta\phi = 1/K$. Crucially, α depends on d_{head} through the discretization ($K = d_{\text{head}}/2$) and is nearly independent of L . The off-diagonal coefficient β has weak power-law decay with L and negligible d_{head} dependence.

The two parameters called τ . Substituting into $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$:

$$\tau_{\text{surr}} = \sqrt{\beta \cdot d_{\text{head}}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}. \quad (35)$$

This fitted-ratio scaling is neither the deployed rule nor evidence for it. The broadband surrogate selects the Cosh family; the separate post-softmax model in Proposition 3.4 balances an $O(\tau^2)$ utility with an $O(\tau^4)$ stiffness and yields

$$\tau_*^2 \propto \frac{d_{\text{eff}}^2}{L}. \quad (36)$$

The equality $c=1$ in the deployed rule is then evaluated by trained-model controls as described in §A.11; it is not obtained from the fitted β/α ratio.

A.13. Stiffness functional derivation

The softmax transport objective in Proposition 3.4 requires choosing a stiffness functional $S(\tau)$ measuring the cost of departing from uniform density. We test a one-parameter family $S_p(\tau) = \frac{1}{d_{\text{head}}} \int_0^1 (\rho_\tau - 1)^2 / \rho_\tau^p d\phi$ using exact (non-Taylor) integrals of the normalized cosh density $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$.

Why χ^2 . Two independent arguments favor $p=1$. (1) *Attention-load asymmetry*: each frequency channel at density $\rho(\phi)$ bears per-channel attention load $\propto 1/\rho$; weighting the squared departure by this load yields $\int (\rho-1)^2 / \rho = \chi^2$. (2) *Discrete-grid effect*: the continuous L^2 integral, evaluated on EVQ's non-uniform cosh-quantile grid, naturally acquires a $1/\rho$ Jacobian factor, producing discrete χ^2 .

Closed form. For $p=1$, the stiffness admits

$$S_{\chi^2}(\tau) = \frac{1}{d_{\text{head}}} \left[\frac{\sinh \tau \cdot \arctan(\sinh \tau)}{\tau^2} - 1 \right], \quad (37)$$

verified numerically to relative error $< 7 \times 10^{-6}$.

Table 6. L -exponent γ (defined by $\tau_* \propto L^{-\gamma}$) under different stiffness functionals, computed via exact numerical optimization over $L \in [128, 4096]$. Target: $\gamma = 0.500$.

Stiffness	p	γ	Note
L^2 norm $\int(\rho-1)^2$	0	0.626	Original choice
Geometric mean	0.5	0.561	
Exponent-matched diagnostic	0.80	0.498	Sensitivity analysis, not used in deployment
Pearson χ^2	1.0	0.465	Channel-load variance penalty (axioms)
KL divergence	—	0.710	Worse than L^2
Reverse KL	—	0.581	
Jeffreys	—	0.621	$\approx L^2$
Rényi ($\alpha=2$)	—	0.807	

Exponent-matched diagnostic. As a sensitivity analysis, one can ask which member of the S_p family would match the target exponent $\tau_* \propto 1/\sqrt{L}$ exactly over this grid. The resulting $S_{\text{derived}}(\tau)$ lies near $p \approx 0.85$ (high-resolution sweep; consistent with 0.80 from the coarser grid), close to but distinct from the Pearson χ^2 choice $p=1$. We do not use this as a derivation of the stiffness; it is reported as sensitivity analysis. The full sweep is reproduced by `scripts/analysis/verify_stiffness_and_regime.py` (pure NumPy, ~ 60 s).

A.14. Effective window L_{eff}^J via softmax-Jacobian Rayleigh quotient

From density perturbation to RoPE frequency shift. The EVQ pure-tether CDF is $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$, with inverse-CDF warp $\Phi_\tau(u) = 1 - \tau^{-1} \text{arsinh}((1-u) \sinh \tau)$. Setting $\theta = \tau^2$ and expanding $\rho_\tau(\phi) = 1 + \theta a(\phi) + O(\theta^2)$ with $a(\phi) = (1-\phi)^2/2 - 1/6$, the quantile displacement takes the closed form

$$\Phi_\tau(u) = u + \tau^2 h(u) + O(\tau^4), \quad h(u) = -\int_0^u a(v) dv = -\frac{1}{6}u(1-u)(2-u), \quad (38)$$

so RoPE frequencies perturb as $\omega_k(\tau) = \omega_k^{(0)}[1 - \tau^2(\log b) h(u_k) + O(\tau^4)]$ with $\omega_k^{(0)} = b^{-u_k}$. Hence $\dot{\omega}_k := \partial_\theta \omega_k|_{\tau=0} = -(\log b) \omega_k^{(0)} h(u_k)$.

Explicit RoPE logit transport direction. Fix a layer/head and query i ; write each 2D RoPE pair as $q_{i,k}^{\mathbb{C}} = q_{i,2k-1} + i q_{i,2k}$, $k_{j,k}^{\mathbb{C}} = k_{j,2k-1} + i k_{j,2k}$, and $\alpha_{ij,k} = q_{i,k}^{\mathbb{C}} \overline{k_{j,k}^{\mathbb{C}}} = C_{ij,k} + i D_{ij,k}$. With relative position r_{ij} and attention scale s_{att} , the RoPE logit is $z_{ij}^{\text{rope}}(\tau) = s_{\text{att}}^{-1} \text{Re} \sum_k \alpha_{ij,k} \exp(ir_{ij} \omega_k(\tau))$. The leading logit transport direction $g_i(j) := \partial_\theta z_{ij}^{\text{rope}}|_{\tau=0}$ has the closed form

$$g_i(j) = \frac{(\log b) r_{ij}}{s_{\text{att}}} \sum_{k=1}^K \omega_k^{(0)} h(u_k) \left[C_{ij,k} \sin(r_{ij} \omega_k^{(0)}) + D_{ij,k} \cos(r_{ij} \omega_k^{(0)}) \right]. \quad (39)$$

This is the Fréchet derivative of the logit with respect to the EVQ allocation perturbation at fixed activations. A finite- τ implementation that avoids first-order phase expansion at large $r_{ij} \omega$ is

$$g_i^{(\tau)}(j) = \frac{1}{s_{\text{att}} \tau^2} \text{Re} \sum_{k=1}^K \alpha_{ij,k} \left[\exp(ir_{ij} b^{-\Phi_\tau(u_k)}) - \exp(ir_{ij} b^{-u_k}) \right], \quad (40)$$

with $g_i(j) = \lim_{\tau \rightarrow 0} g_i^{(\tau)}(j)$.

Primary definition: Rayleigh quotient of the softmax Jacobian. For the trained attention distribution $p_i(j)$ on keys $j \in \mathcal{K}_i$, $J_{\text{sm}}(p_i) = \text{diag}(p_i) - p_i p_i^\top$ satisfies $g_i^\top J_{\text{sm}}(p_i) g_i = \text{Var}_{j \sim p_i}[g_i(j)] = \frac{1}{2} \sum_{j,k} p_i(j) p_i(k) (g_i(j) - g_i(k))^2$. Since $J_{\text{sm}}(p_i) \mathbf{1} = 0$, we center g_i by the key-wise mean operator P_i . The effective length is then defined by the *ratio-of-means*

$$\frac{1}{L_{\text{eff}}^J} = \kappa_{\text{att}} = \frac{\mathbb{E}_{\ell,h,i} [g_i^\top J_{\text{sm}}(p_i) g_i]}{\mathbb{E}_{\ell,h,i} [\|P_i g_i\|_2^2]}, \quad (41)$$

This is the directional curvature of the *schedule-change* KL: for $p_\epsilon = \text{softmax}(z + \epsilon g)$, $D_{\text{KL}}(p \| p_\epsilon) = \frac{1}{2} \epsilon^2 g^\top J_{\text{sm}}(p) g + O(\epsilon^3)$, and for diffuse $p=1/L$ the quotient reduces to $1/L$. The phase-variance utility Q_1 in Proposition 3.4 is a distinct

pairwise positional functional. Replacing its $1/L$ factor by $1/L_{\text{eff}}^J$ is therefore a possible modelling extension, not a consequence of the KL identity and not an empirical result used by the paper.

A.15. Pure-tether branch: Fisher-forcing coefficient and residual bound

Setup. The full stationary ODE under $\mathcal{C}_{\text{app}}$ plus a Fisher utility is $\rho'' - \tau^2 \rho = \gamma b^{-2\phi}$. Writing $s = 2 \log b$ and $q_s(\phi) = e^{-s\phi}$, the pure-tether branch $\rho_{\text{EVQ}}(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$ solves the homogeneous problem; the full solution is $\rho_{\text{full}} = \rho_{\text{EVQ}} + \delta \rho_F$.

Explicit form of $\gamma(b)$. The Fisher utility in the Euler equation $\alpha \rho + \beta(G\rho) - \lambda_F V_b + \nu = 0$ has $V_b(\phi) = \eta_F(\phi) b^{-2\phi}$, where $\eta_F(\phi)$ is the activation-conditioned Fisher coefficient with

$$\eta_F(\phi_k) = \frac{1}{2 s_{\text{att}}^2} \mathbb{E}_{\ell, h, i, j} [w_{\ell h i j} |\alpha_{\ell h i j, k}|^2], \quad \alpha_{\ell h i j, k} = q_{i, k}^{\mathbb{C}} \overline{k_{j, k}^{\mathbb{C}}}. \quad (42)$$

Using $(G\rho)'' = -\rho$, differentiating the Euler equation twice gives $\rho'' - \tau^2 \rho = (\lambda_F / \alpha) V_b''$. For slowly varying η_F compared with $e^{-s\phi}$,

$$\gamma(b) = \frac{\lambda_F}{\alpha} \bar{\eta}_F s^2 \left[1 + O\left(\frac{|\eta_F'|}{s \bar{\eta}_F} + \frac{|\eta_F''|}{s^2 \bar{\eta}_F}\right) \right], \quad s = 2 \log b. \quad (43)$$

The full $\eta_F(\phi)$ -local version is $\gamma_{\text{loc}}(\phi; b) = (\lambda_F / \alpha) [\eta_F''(\phi) - 2s \eta_F'(\phi) + s^2 \eta_F(\phi)]$.

Suppression is L^1 /CDF, not pointwise. The particular solution $\delta \rho_F^{\text{part}} = \gamma(b) q_s / (s^2 - \tau^2)$ combined with (43) gives pointwise amplitude

$$\delta \rho_F^{\text{part}}(\phi) \approx \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{s^2 - \tau^2} e^{-s\phi}, \quad (44)$$

so the $(s^2 - \tau^2)^{-1}$ factor is cancelled by the s^2 in $\gamma(b)$, and $\delta \rho_F^{\text{part}}$ is *not* pointwise small in s . The true suppression comes from the boundary-layer width of q_s : $\|q_s\|_1 = (1 - e^{-s})/s = O(1/\log b)$ and $\|q_s\|_2 = O(1/\sqrt{\log b})$. Let $C_{\text{bc}}^{(p)}(\tau)$ denote the finite L^p operator norm of the homogeneous correction enforcing the boundary and mass constraints, away from resonance. Then

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_1 \leq C_{\text{bc}}^{(1)}(\tau) \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{|s^2 - \tau^2|} \frac{1 - e^{-s}}{s} = O\left(\frac{1}{\log b}\right), \quad (45)$$

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_2 \leq C_{\text{bc}}^{(2)}(\tau) \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{|s^2 - \tau^2|} \sqrt{\frac{1 - e^{-2s}}{2s}} = O\left(\frac{1}{\sqrt{\log b}}\right), \quad (46)$$

and the CDF-level error is $\|F_{\text{full}} - F_{\text{EVQ}}\|_{\infty} \leq \|\delta \rho_F\|_1 = O(1/\log b)$. For fixed Fisher coefficient and away from resonance, forcing is therefore small in CDF / L^1 transport at large b , but not necessarily pointwise small near $\phi=0$.

Warp amplification at large τ . Via the inverse-CDF map,

$$\|\phi_{\text{full}} - \phi_{\text{EVQ}}\|_{\infty} \leq \frac{\|\delta \rho_F\|_1}{\inf_{\phi} \rho_{\text{full}}(\phi)}, \quad \inf_{\phi} \rho_{\text{EVQ}}(\phi) = \frac{\tau}{\sinh \tau}, \quad (47)$$

so, when the residual is small enough that ρ_{full} remains positive and comparable to ρ_{EVQ} , CDF perturbations can be amplified at the scale $\sinh \tau / \tau$. The bound does not justify omitting the forcing branch at large τ .

Dimensionless diagnostic. Using $\|\rho_{\text{EVQ}} - 1\|_2 = \tau^2 / \sqrt{45} + O(\tau^4)$ and $\Gamma_b := (\lambda_F \bar{\eta}_F / \alpha) s^2 / (|s^2 - \tau^2| \tau^2)$, the L^2 -level ratio is

$$R_F := \frac{\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_2}{\|\rho_{\text{EVQ}} - 1\|_2} \leq C_{\text{bc}}^{(2)} \Gamma_b \sqrt{45} \sqrt{\frac{1 - e^{-2s}}{2s}} = O\left(\frac{1}{\sqrt{\log b}}\right). \quad (48)$$

Second-order utility deficit. Let A_{τ} be the constrained Hessian of the surrogate on zero-mass perturbations and let $q_s^{\perp}$ be the projection of q_s onto that tangent space. The full branch improves the quadratic objective by

$$\Delta_F = \frac{1}{2} \gamma(b)^2 \langle q_s^{\perp}, A_{\tau}^{-1} q_s^{\perp} \rangle, \quad (49)$$

so dropping the forcing term is $O(\gamma(b)^2)$: if $R_F \leq r$, the missed correction is $O(r^2)$ relative to the tether deformation energy.

A.16. Channel-load motivation for Pearson χ^2

Normalization convention. Throughout this paper, $S_{\chi^2}(\tau)$ denotes the d_{head} -normalized stiffness $S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 (1 - \rho_\tau)^2 / \rho_\tau d\phi$, with leading expansion $\tau^4 / (45 d_{\text{head}}) + O(\tau^6)$ consistent with Proposition 3.4. Identities below that produce a bare $\tau^4 / 45$ leading coefficient apply to $d_{\text{head}} S_{\chi^2}(\tau)$ (the un-normalized integral); the d_{head} factor enters via the surrogate diagonal $\alpha = 1/d_{\text{rot}} (= 1/d_{\text{head}}$ for MHA via §A.12; $= 1/d_{\text{rot}}$ for compressed-RoPE MLA via §G.1).

Channel-load axioms. Inverse-CDF quantization places channels with cell width $\ell(\phi) = 1/\rho(\phi)$ and samples channels according to $dP_\rho = \rho d\phi$. Under the modeling axioms (i) channel-measure averaging, (ii) load ℓ as the primitive local quantity, (iii) second central moment, (iv) uniform density has zero stiffness, (v) local additivity, the canonical load-variance penalty is

$$S(\rho) = \text{Var}_{\phi \sim \rho d\phi} \left[\frac{1}{\rho(\phi)} \right] = \int_0^1 \frac{(1 - \rho(\phi))^2}{\rho(\phi)} d\phi = S_{\chi^2}(\rho). \quad (50)$$

We treat this as a canonical motivated choice under the stated axioms, not a uniqueness claim.

Load-moment exponent. Distinct from the weighted- L^2 family $S_p(\tau) = \frac{1}{d_{\text{head}}} \int (\rho_\tau - 1)^2 / \rho_\tau^p d\phi$ used in Table 6 (where Pearson χ^2 is $p=1$), we also examine a load-moment family $S_p^{\text{mom}}(\rho) = \mathbb{E}_\rho[(1/\rho - 1)^p]$. Substituting the EVQ cosh density and minimizing $S_p^{\text{mom}} - \lambda U$ gives $\tau^{2(p-1)} \asymp d_{\text{head}}^2 / L_{\text{eff}}^J$, hence $\tau^* \propto (L_{\text{eff}}^J)^{-1/(2p-2)}$; only $p=2$ yields the observed $L^{-1/2}$ exponent within this family. The two families parametrize different f -divergence neighborhoods and need not match on the finite- τ grid.

f -divergences at finite τ . Smooth f -divergences (KL, Jeffreys, Rényi) are locally quadratic around $\rho=1$. Over the finite grid used in Table 6, however, their fitted exponents range from 0.58 to 0.81. This is model sensitivity, not a trained-PPL comparison.

A.17. Discrete-channel transport gap and the MLA channel-scarcity mechanism

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^\phi \rho$, $Q(u) = F^{-1}(u)$, midpoint grid $u_k = (k - \frac{1}{2})/K$, and $\phi_k = Q(u_k)$. The empirical channel measure is $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. Direct $\|\rho_K - \rho\|$ is ill-posed: one cannot subtract an atomic measure from a density. We therefore work in Wasserstein distance for transport and in a quantile-cell histogram ρ_K for density.

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (51)$$

Defining the quantile-cell histogram $\rho_K(\phi) = 1/(K|I_k|)$ on $I_k = [Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (52)$$

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$, $m_\tau = \tau / \sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (53)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km} + O(K^{-2}). \quad (54)$$

The singular δ -component of K_{app} must be interpreted as a cell average (not as a literal atom-kernel pairing), after which the bound applies to the smoothed kernel.

High-resolution distortion. For a smooth weighted distortion w ,

$$\mathcal{D}_K[\rho] \approx \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (55)$$

This is the finite-channel mechanism: a shaped ρ can reduce the weighted inverse-density load $\int w/\rho^2$; the absolute benefit scales as K^{-2} for smooth quantization error and as K^{-1} for transport/kernel terms.

Channel scarcity amplifies the density-shape correction. The standard MHA configuration has $d_{\text{head}}=d_{\text{rope}}=64$ and $K=32$; the MLA study has $d_{\text{head}}=64$, $d_{\text{rope}}=32$, $d_{\text{nope}}=32$ and $K=16$. Halving K scales the K^{-2} component of the distortion bound by $4\times$ and the K^{-1} component by $2\times$. The observed MLA gain (§4.5) is a test of the predicted direction, not a controlled quantitative test of either bound.

Caveat. At large τ , $m_\tau=\tau/\sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds can therefore become vacuous in the same regime where the Fisher-residual analysis no longer guarantees a small warp error (§A.15).

B. Experimental Details

B.1. Scope of this appendix

The exact-range and mature-model primary protocols are documented separately in Apps. C and F. This section records the submitted scratch-model and cross-modal experiments that remain supporting evidence for the substrate-range and cross-modal scope claims. The learned-frequency comparator appears in App. E.

B.2. Reproducibility snapshot

Table 7. Compact reproducibility summary for supporting scratch-model experiments.

Setting	Train length	Scale	Seeds	Main datasets
EVQ×RAMP	2048	454M	3 + 3	FineWeb-Edu + 10% passkey mix
MLA	8192	432M	3	FineWeb-Edu

Table 8 lists the training hyperparameters shared across the supporting text experiments.

Table 8. Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 3×10^{-4} (125M), 2×10^{-4} (454M/MLA/750M)
LR schedule	Cosine decay to $0.1\times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 16 (125M), 2–4 (454M/750M), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

Within each text comparison, the architecture, optimizer, data order, and training loop are fixed, and the only intervention is the initialized inverse-frequency table. The scratch-text experiments use base $b=500\text{K}$ unless stated otherwise.

Table 9. Single-seed 750M continued-pretraining check (2K $\rightarrow$ 4K, 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials. This supporting row is separate from the multi-seed primary controls.

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Strict AR exact @8K	0%	77.5%
Global strict AR exact	66.67%	92.5%

B.3. Larger-scale supporting evidence

The 750M ‘continue@4K’ experiment starts both arms from the same 2K geometric checkpoint and applies a 500M-token continuation with either the geometric or EVQ table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, strict autoregressive exact match is 0% versus 77.5%. This is a single-seed post-hoc continuation result, not a from-scratch allocation control.

B.4. Modality-dependent τ correction for video DiT

The text-style reference used for this sweep gives $\tau = K_t / \sqrt{T} = 2.83$ for $K_t=16$ and $T=32$, whereas the best tested Video DiT point is $\tau=1.5$ ($0.53\times$). We use this empirical correction only to select the supporting video configuration; the sweep does not derive a modality-general coefficient.

B.5. Cross-modal evidence: video DiT with bidirectional attention

We evaluate a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames ($4\times$) with temporal RAMP. For each seed, the geometric and EVQ-Cosh ($\tau=1.5$) branches are trained in the same head-to-head run.

Table 10 reports two independent seeds. The within-seed contrasts favor EVQ on train, all-frame extrapolation, and far-frame extrapolation MSE in both runs; the two-seed mean deltas are -26% , -16% , and -33% . This is a cross-modal scope check, not an estimate of a general video effect size.

Table 10. DiT head-to-head evaluation (129.6M, bidirectional attention, 3D RoPE). Each seed is one shared-run comparison; values are denoising MSE at $t=0.5$.

Seed	Metric (RAMP)	Geo ($\tau=0$)	EVQ ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	-21%
42	All extrap	0.00724	0.00606	-16%
42	Far extrap	0.00989	0.00639	-35%
137	Train MSE	0.01113	0.00761	-32%
137	All extrap	0.00713	0.00598	-16%
137	Far extrap	0.00856	0.00601	-30%
Mean Δ		$-26\% / -16\% / -33\%$		

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. In the pre-specified diagnostic, $\tau=1.2$ leaves 11/16 channels above this threshold (effective rank 9), whereas $\tau=1.5$ leaves 12/16 (rank 10). Repeating the head-to-head comparison at base=1,000 places 13/16 geometric channels above the threshold. In this single-seed run, $\tau=1.2$ and $\tau=1.5$ differ by less than 0.3% on the reported metrics and both improve far-frame MSE by about 48% relative to Geo. This is consistent with the diagnostic; it is not an equivalence or causal estimate.

Table 11. Base = 1,000 DiT head-to-head (129.6M, seed 42, RAMP). The two EVQ settings are within 0.3% on the reported metrics in this run.

τ	Train MSE	All extrap	Far extrap	Δ_{far} vs Geo
0.0 (Geo)	0.02037	0.01274	0.01646	—
1.2	0.00933	0.00820	0.00859	−48%
1.5	0.00930	0.00822	0.00858	−48%

C. Identification protocol and provenance

C.1. Exact-range control at 151.9M

What is pinned. The control is designed so that every scalar quantity a base/range method can express is identical between arms. Within each seed pair the two arms share: model architecture (151,898,880 parameters); trainable initialisation; training length 256; token order; optimiser, LR schedule, global and micro batch; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2 = 30$ interior frequencies.

Arms. *Baseline:* a paper-faithful reimplement of the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026), at training base 256. We did not identify a public author implementation when implementing the control, so the comparison is against the rule as published, not an official-code reproduction. *Intervention:* endpoint-normalised cosh interior spacing at $\tau=4$. Note that the endpoint-normalised cosh arm is *not* identical to the deployed midpoint EVQ-COSH grid of Eq. (4), whose τ also shifts the sampled extrema; endpoint normalisation is what makes the exact-range control possible, and this is a shape diagnostic rather than a deployment configuration.

Evaluation. Two conditions. *fixed:* each checkpoint retains its training range. *target-matched:* the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$.

Seed-42 absolute values. Under *fixed* range, PPL is 28.386/142.851/342.056/564.353 (uniform) against 29.331/88.615/278.659/504.131 (cosh) at 256/512/1,024/2,048. Under *target-matched* range, 28.386/28.038/35.209/100.111 (uniform) against 29.331/29.804/42.230/132.271. Cosh wins 32/32, 27/32, 22/32 anchors at 512/1K/2K under *fixed*, and only 5/32, 1/32, 2/32 under *target-matched*.

Reading. The *fixed* column identifies the optimisation variable: interior allocation is not reducible to scalar base/range selection. The *target-matched* column bounds the deployment claim: when range transport is available to both arms, target-aware retargeting is the stronger lever in this protocol. Both are reported in the body (§4.2).

C.2. Exact-range factorial at 50.9M

Design. The schedule arms, including the $1.25\times$ multiplier, were specified before any result was inspected. Grid: 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 structural configurations; 5 schedules (Geo, anchored cosh at $0.75\times$, $1.00\times$ [the formula point], $1.25\times$, and a deformation-matched exponential); 3 seeds. Main factorial 180/180 completed, boundary- τ follow-up 12/12 completed, covering the complete pre-specified matrix.

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$, never an arithmetic mean of raw PPL.

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); cosh $0.75\times$ 6.202858 (494.16); cosh $1.00\times$ 6.202094 (493.78); cosh $1.25\times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5\times$ arm is better; at formula $\tau=8$ the $0.75\times$ arm is. Optima move inward from both ends, which bounds the formula’s implied range but does not yield a replacement closed-form rule.

Scope limit. This is a short-budget 50.9M mechanistic NLL control trained on a deterministically repeated local WikiText-2 raw stream — not a large-scale pre-training distribution, and a different corpus from the 151.9M control and from the from-scratch runs. It is not scale evidence and not capability evidence, and its effect size is not comparable to the 151.9M single-configuration contrast: the factorial establishes cross-configuration *direction*, whereas the 151.9M control provides the effect size within its configuration.

D. The fixed-index ramp scaler

Every arm marked +RAMP in this paper uses our fixed-index smooth-ramp range scaler from the NTK-by-parts family that includes YaRN (Peng et al., 2024). It is used as a fixed-scale substrate control, not as a reproduction or tuning study of reference YaRN.

Its role is a fixed-scale substrate control: it is held *identical* across the arms being compared — same implementation, same fixed cut indices, same scale s — so that the only difference between Geo + RAMP and EVQ-COSH + RAMP is the training-time table each substrate presents to the same operator. It therefore measures substrate-dependent leverage at the fixed tested scale, rather than tuned range-method performance.

Implemented components. For K frequency channels, the scaler uses fixed channel-index cut points $k_{lo} = \lfloor 0.20K \rfloor$ and $k_{hi} = \lfloor 0.90K \rfloor$. Between them it applies the cubic smoothstep $r(x) = x^2(3 - 2x)$ and replaces each inverse frequency ω_k by

$$\omega_k / \left(s^{r_k} t^{r_k/2} \right), \quad t = 1 + 0.07 \log_2 s.$$

Thus an interpolation ramp is present, but its boundaries are fixed by channel index rather than YaRN’s wavelength-derived $\beta_{fast}/\beta_{slow}$ rule. The factor t is folded into frequency scaling; there is no separate attention-logit magnitude scaling. These differences are why we label the operator RAMP rather than YaRN.

E. Learned-frequency comparator

Table 12. Learned-frequency comparator (125M, $L_{train}=128$, FineWeb-Edu, $128 \rightarrow 8K$). Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). **We rest no allocation-shape attribution on this table (§4.4):** the learned-inv-freq arm carries 32 trainable parameters, so any gap confounds allocation shape with parameterisation and optimisation effort. The arm is a layer-shared learnable inverse-frequency baseline — one learned scalar per band, shared across layers and heads — and therefore belongs to the learned-table family (Karypis et al., 2026) rather than the learned positional-operator family (Zheng et al., 2024).

Method (128 $\rightarrow$ 8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 $\pm$ 1.3	437.9 $\pm$ 12.2	−14.8%
Learned inv-freq (layer-shared)	32	183.6	455.3	−11.4%
same, 10 $\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	−35.0%

The 32-parameter arm in Table 12 is a layer-shared learnable inverse-frequency baseline: one learned scalar per frequency band, shared across layers and heads. It received a dedicated positional-parameter learning-rate sweep at 10 $\times$ and 100 $\times$ the base rate, and the stronger 100 $\times$ setting is the 455.3 row. This answers a tuning-budget question but not the attribution question: a learned-parameter arm cannot isolate allocation shape from parameterisation, which is why §4.2 uses fixed zero-parameter schedules under an identical operator instead, and why no allocation-shape claim in this paper depends on this table.

On method identity: DAPE (Zheng et al., 2024) is a learned positional *operator*, whereas this comparator learns inverse frequencies directly and belongs to the learned-table family later studied at scale by Karypis et al. (2026). The distinction matters for what the row can be read as evidence for. Read as a learned-table comparison, its direction is consistent with the independent finding that learned tables degrade more sharply than standard RoPE under naive extrapolation.

Table 13. **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on; LLaMA uses matched Q/K/V/O LoRA. Training and evaluation rows are disjoint but share the 13 generator families: task-adapted length transfer, not unseen-task transfer.

Model / protocol	Arm / metric	1×	2×	4×
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official macro	82.16	0.08	0
	EVQ-COSH, official macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official macro	72.19	2.02	0.38
	EVQ-COSH, official macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official macro	94.44	0.295	—
	EVQ-COSH, official macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

F. Mature-scale protocols

F.1. OLMo-2 1.485B

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (OLMo Team, 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, but explicit position IDs expose target-range relative phases. This is therefore a physical-memory cap, not phase-free 4K training. On 8K `niah_single_1`, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set. Natural-text NLL and the distinct autoregressive endpoint are reported separately in Table 3.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start independently from the matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length, Native/EVQ-COSH exact match is 22.0/21.5% at 4K, 0/17.5% at 8K, and 0/4.0% at 16K. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length. Thus this is held-out task-family length transfer, not an unmodified LongBench leaderboard result. The RULER adapters are independent of the QA adapters: Table 13 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families.

Family-level in-window structure. For the selective-Q/K RULER adapters at 4K, the official-score changes (EVQ-COSH minus Native) for `niah_single_1/2/3` are −5/0/−60 percentage points, averaging −21.67. The changes for `niah_multikey_1/2/3`, `niah_multivalue`, and `niah_multiquery` are −65/−95/−50/−26.25/−25, averaging −52.25. This split is a post-hoc mechanism diagnostic over one matched seed, not a separate benchmark claim; it motivates the multi-reference prediction in §5.

From initialisation. Starting from the public step-0 initialisation, one 2.097B-token EVQ-COSH trajectory reaches PPL 167.45/156.87/159.64 against the released Geo step-1,000 checkpoint’s 161.19/163.88/182.73 at 4/8/16K. The paired EVQ-COSH-minus-Geo NLL is +0.0381/−0.0437/−0.1351, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K. This is one same-initialisation, same-scientific-recipe trajectory: EVQ-COSH used a reviewed single-GPU HF loop, whereas the released Geo checkpoint used the original distributed trainer. This row evaluates natural-text LM; capability is evaluated by the separate autoregressive protocols above.

F.2. LLaMA-3-8B-Instruct

Probability and causal source use. In the matched seed-42, 300-step Q/K/V/O LoRA protocol, EVQ-COSH-minus-Native NLL is $+0.390 / -1.510 / -2.048$ at 8/16/32K. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Removing the remote gold block from every head changes NLL by -0.0095 for Native and $+1.5055$ for EVQ-COSH. These are matched teacher-forced probability and causal-routing measurements; autoregressive task accuracy is evaluated separately.

RULER-family continuation. A separate 516-step continuation uses identical physical-8K supervision over all 13 RULER families, with one seed per arm and disjoint generated training/evaluation rows. At 16K, Native/EVQ-COSH official macro is 0.295/14.03% and normalized exact is 0/1.54%. At the 8K training length the metrics disagree, so Table 13 reports both official macro (94.44/77.60%) and normalized exact (17.69/21.54%).

Endpoint scope. NLL, attention routing, strict autoregressive exact match, multi-hop QA, and RULER remain separate endpoints. The 2Wiki and RULER continuations use the same task families at train and test with disjoint rows, so they establish task-adapted length transfer, not unseen-task transfer. The natural-text and routing rows make no task-transfer claim.

G. Supporting Results

G.1. Multi-head Latent Attention

We test the finite-budget hypothesis in a 432M-parameter MLA transformer with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ uses the pre-specified $\tau=1.414$ setting. This experiment tests the frequency allocation at fixed architecture; it does not identify a general MLA operating rule for τ .

Table 14. MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). The legacy scaler is this run’s wavelength-blend operator, not the RAMP used elsewhere.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
GEO	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
GEO+legacy(s=4)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ+legacy(s=4)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

EVQ reduces 16K PPL by 31.1% (138.8 $\rightarrow$ 95.6), while 8K PPL changes by $+0.9\%$. All three training seeds favor EVQ at 16K. EVQ also improves on GEO+legacy at 16K and ties it to printed precision at 24K; EVQ+legacy has the lowest mean PPL at every reported length. Because the legacy operator is not a matched-range control, these rows establish only the ordering in this architecture and protocol.

H. Supporting Experiments (from Main Text)

The following results record additional submitted protocols with their seed scope.

Table 15. $L=256$ RAMP composition study (125M, 3-seed mean PPL). Among the tested settings, $\tau=4.0$ has the lowest mean PPL at every reported extrapolation ratio.

Method ($L=256$)	$4\times$	$8\times$	$16\times$	$32\times$
Geo+RAMP	60.7	97.5	190.4	260.2
EVQ-COSH ($\tau=2$)+RAMP	51.6	66.7	107.9	141.7
EVQ-COSH ($\tau=4$)+RAMP	50.9	60.9	84.1	99.6

Cross-modal scope. In two 129.6M DiT shared-run comparisons, the EVQ branch has lower denoising MSE on train, all extrapolation frames, and far extrapolation frames in both seeds. Full values and the one-seed base diagnostic are in §B.5.